

**2024 年青年科学家长三角国际科研实
习项目“GripS”结题报告**

**2024 Global Research Immersion
Program for Young Scientists
Final Report**

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实 验 室 Lab Name Centre for Neural and Intelligent
Engineering

科研导师 PI Name Shouyan Wang

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复旦大学制表

Fudan University

一、基本信息 Basic Information

承担高校

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复旦大学 Fudan University

实习人

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Information

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导师简介

PI Introduction

Professor Shouyan Wang received the BSc degree in Biomedical Engineering in 1994, and the MSc and PhD degree in Physiology in Xi'an, China, in 1997 and 2000, respectively. He was a Postdoctoral Research Fellow in the Functional Neurosurgery Department of JR Hospital and Department of Physiology, Anatomy and Genetics at University of Oxford from 2002 to 2007. He worked as a Lecturer in the institute of Sound and Vibration Research at University of Southampton from 2007 to 2012. He became a Professor at Suzhou Institute of Biomedical Engineering and Technology of Chinese Academy of Sciences, and the Directors of the Biomedical Electronics Department, Key Lab of Neural Engineering and Technology at Suzhou from 2012 to 2017. He joined Fudan University in 2017, and has been the Vice Dean of the Institute of Science and Technology for Brain-inspired Intelligence (ISTBI) and Director of Neural and Intelligent Engineering Centre at ISTBI. Professor Wang is the executive directors of National 111 International Cooperation Center for Computational Neuroscience and Brain-inspired Intelligence and International Cooperation and Innovation Hub of China Association for Science and Technology.

His research focuses on the intelligent neuromodulation of deep brain stimulation for neurological diseases, including identification of biomarkers from human deep brain local field potentials, technology development of miniaturized adaptive electrical or optical stimulator, and monitoring of motor or sensory behaviors with wearables devices. Professor Shouyan Wang works toward integrating engineering, neuroscience, neurology and neurosurgery to advance the translational research from deep brain stimulation technology to neurological disease treatment. Professor Wang is the chairman of Neuromodulation Branch of Chinese Neuroscience Society.

实验室

Lab Information

名称

Name

神经与智能工程中心 Centre for Neural and Intelligent Engineering

所属院系

Department

类脑智能科学与技术研究院

实验室简介

Lab Introduction

To improve the clinical treatment of some key neurological and psychiatric disorders, the Centre aims to develop the neural sensing and processing approaches to neuro-dynamics information, and explore the integration strategy of brain information and plasticity mechanisms of brain functions. By combining neural coding and intelligent theories, the Centre will develop neural sensing, neural modulating, neural computing technologies, and innovation in brain-intelligence technology. It will also build intelligent systems on-chip, and create a self-learning intelligent human-machine interface. Through research in intelligent neuro-modulation and neural-bionics, our goals are to provide intelligent devices and creative solutions to protect, improve and modulate the human brain .

个人实习感言

Internship Summary

During this internship I was collaborating with current Fudan students on their Protein Function Prediction Machine Learning project, evaluating the capacity to predict function from protein structure.

结题报告 Final Report

说明：请简述本次实验室科研实习的相关研究经历及研究结果

Please briefly describe the research experience and achievements related to this laboratory research internship.

Protein Function Prediction (PFP) is emerging as one of the most transformative areas of research today, given its profound implications for science and medicine.

Currently, determining a protein's structure, chemical properties, and molecular and biological functions is an incredibly resource-intensive process, often requiring years of research and investments amounting to millions of dollars. Despite these efforts, the largest digital protein database currently contains only about 500,000 fully labeled, studied, and reviewed proteins. In stark contrast, there are approximately 250 million unreviewed, unlabeled, and unknown proteins—500 times more unknown proteins than known ones.

Enter the fields of bioinformatics and machine learning. These disciplines have the potential to revolutionize biochemistry research by significantly increasing efficiency, which could accelerate the discovery of new medicinal treatments. Automating the identification of potential candidate proteins for drug targeting holds immense promise for benefiting humankind. This is because all organic life is fundamentally based on proteins performing autonomous functions according to their specific designs. Proteins are composed of amino acids, and their sequence determines their structure, bonding, and, consequently, their function.

Historically, some of the most groundbreaking protein discoveries occurred serendipitously through common sense observations. For example, Taq polymerase, a heat-stable DNA polymerase derived from the bacterium *Thermus aquaticus*, that lives by the hot sea vents, is essential for the Polymerase Chain Reaction (PCR) technique, which revolutionized genetic research. Similarly, Channelrhodopsins, light-sensitive Na⁺

ion channels, have been instrumental in neuroscience since 2005, enabling a shift from correlational to causal research in cognitive neuroscience. The impact of these proteins on scientific research is undeniable. Now, consider the untapped potential of the 250 million unstudied proteins, waiting to be discovered and utilized.

Given this potential, major technology companies have recognized the value of this research area and have established dedicated departments to advance it. Meta (formerly Facebook) developed the ESM-2 model on Hugging Face, a pretrained large language model based on the BERT architecture, which outperforms previous models in predicting protein structure and function.

Another notable model is AlphaFold 2 by Google, which, although not used in this particular research project, remains highly relevant. AlphaFold 2 is currently the most accurate model for structural prediction, achieving 76% accuracy for ligand-ligand interactions and 62% for protein-protein interactions. However, while ESM-2 is open-source and publicly available for integration into research, Google's AlphaFold is accessible only on a limited basis through their servers, with no open-source code available. While the advancements driven by these investments are commendable, the decision to keep some models closed-source potentially hampers the collaborative progress that could drive us closer to achieving a 90% accuracy threshold—high enough to justify significant investment in the biological research of identified target proteins.

The development of sophisticated computational methods has been central to advancements in PFP. The Gene Ontology (GO) system, for example, classifies proteins into hierarchically related functional classes organized into three different ontologies: Molecular Function (MF), Biological Process (BP), and Cellular Component (CC). These classifications describe different aspects of protein functions and aid in their prediction. The Naive Algorithm is another widely used method for protein function prediction, leveraging the functions of homologous proteins with similar sequences.

Deep learning has also led to unprecedented improvements in protein function prediction. Graph Convolutional Networks (GCNs), such as DeepFRI, have been used to predict protein functions by leveraging sequence features extracted from protein language models and protein structures. DeepFRI, a GCN-based method, has outperformed current leading methods and sequence-based Convolutional Neural Networks (CNNs) and scales to the size of current sequence repositories.

Other methods include the widely used BLAST (Basic Local Alignment Search Tool), which compares a query sequence to a database of known sequences to identify homologs with known functions. The k-nearest neighbors algorithm (KNN) is another method that predicts the function of a protein based on the functions of its closest neighbors. Net-KNN, a variant of KNN that uses a neural network, has been shown to outperform the traditional KNN algorithm in some cases.

This lab's objective was to evaluate and compare these PFP methods for efficiency in predicting function. The conclusion was that the DNN-EMS2 method largely outperformed the other methods, however an ensemble method of combining the consensus from all methods was overall the best at being able to predict the function in the test data.

Just getting the code to function took the longest amount of time but this project gave me great insight into how machine learning can be applied to the field of bioinformatics whereas in the past I only had experience using ML for neuroimaging.

I found this experience very rewarding and enjoyable, working with an international group of students, and interacting with the students here at Fudan to get a better idea of what type of work is going on here.

导师评语 PI Evaluation:

Alban Malaj has shown exceptional dedication and enthusiasm throughout the summer school program.

考核情况 Assessment:

考核形式 Form of Assessment :

Attendance 10% Classroom Performance 30% Course Report 60%

(2) 考核结论 Result of Assessment :

☒优秀 Excellent ☐合格 Passed ☐不合格 Failed

导师签名 PI Signature :

日期 Date : 2024.8.9

